AS · Unit 2 · WCH12 | Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Energetics

Every chemical reaction takes in or gives out energy, because bonds have to be broken and new ones made. This topic shows you how to define, measure and calculate these enthalpy changes: with a thermometer and a polystyrene cup, with Hess cycles that find values nobody can measure directly, and with bond enthalpies.

SPEC 6.1 Enthalpy change and standard conditions

WHAT THIS SPEC POINT ASKS

Know that ΔH is the heat energy change measured at constant pressure, and that standard conditions are 100 kPa and a stated temperature, usually 298 K.

What ΔH means

- **Enthalpy (H)** is the heat energy stored in a substance. You can't measure it directly, but you can measure the **change**, ΔH, when a reaction happens.
- **ΔH is the heat energy change measured at constant pressure.** A reaction in an open beaker or a polystyrene cup is at constant (atmospheric) pressure, so the heat it gives out or takes in is ΔH.
- Units: **kJ mol^{-1}**, i.e. per mole of the reaction as written in the equation (or per mole of the named substance in a definition).
- **System** = the reacting chemicals. **Surroundings** = everything else: the water or solution, the container, the air. A thermometer measures the surroundings.

Standard conditions

Condition	Value
Pressure	**100 kPa** (10^5 Pa)
Temperature	a stated temperature, **usually 298 K** (25 °C)
Solutions	1 mol dm^{-3}
Substances	in their **standard states**: their normal physical state at 100 kPa and 298 K (e.g. $H_2O(l)$, C(s, graphite), $Br_2(l)$, $O_2(g)$)

- A standard enthalpy change is written with the symbol $^{\ominus}$, e.g. $\Delta H^{\ominus}_{298}$. The temperature is "usually" 298 K because data can be quoted at other temperatures as long as the temperature is stated.
- **State symbols matter.** $H_2O(l)$ and $H_2O(g)$ have different enthalpies: condensing steam releases heat. So ΔH for a reaction making liquid water is more negative than one making steam. Always write state symbols in enthalpy equations.

MARK-SCHEME WORDING

- Standard conditions: "298 K / 25 °C" and "100 kPa / 1 atm / 101 kPa / 1 × 10^5 Pa". ACCEPT "a specified temperature".
- DO NOT ACCEPT "298 °K", or a wrong unit such as "101 Pa". "Room temperature" alone is ignored.
- Final answers to enthalpy calculations need a **sign and units** (kJ mol^{-1}) for the last mark.

Often asked as: part of a definition ("…under standard conditions, which are…"), or the sign-and-units mark at the end of a calculation.

COMMON MISTAKE

Writing "standard conditions" when you mean "standard **states**". Conditions are pressure and temperature; states are the physical states of the substances (WCH12 Jan 2024 examiner report).

Leaving out state symbols in an enthalpy equation.

SPEC 6.2 Sign convention: exothermic and endothermic

WHAT THIS SPEC POINT ASKS

Know that exothermic reactions have a negative ΔH and endothermic reactions a positive ΔH.

	Exothermic	Endothermic
Heat	given **out** to the surroundings	taken **in** from the surroundings
Temperature of surroundings	**rises**	**falls**
Products compared with reactants	**lower** enthalpy	**higher** enthalpy
Sign of ΔH	**negative** (ΔH < 0)	**positive** (ΔH > 0)
Examples	combustion, neutralisation, most oxidation, respiration, displacement	thermal decomposition (e.g. $CaCO_3$), photosynthesis, dissolving NH_4NO_3, $KHCO_3$ + HCl

- The sign describes the **system**. When the system loses heat, its enthalpy falls, so ΔH is negative, even though the thermometer reading goes up.
- Bond breaking is endothermic; bond making is exothermic (see 6.9). Whether the overall reaction is exo- or endothermic depends on which is bigger.

EXAM TIP

Use the experiment to check your sign before you write the answer. **Temperature went up → ΔH negative.** Temperature went down → ΔH positive. A thermal decomposition must come out positive; if yours is negative, recheck the cycle.

MARK-SCHEME WORDING

- "Heat energy is **released** (to the surroundings)" for exothermic; "heat energy is **absorbed** / taken in" for endothermic.
- The final calculation mark is usually "answer with **correct sign** and units".

Often asked as: the sign of a calculated ΔH, or "explain why this reaction is exothermic" from a temperature change.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

A temperature rise means the reaction is exothermic, so $\Delta_r H$ must be given as negative even when the question doesn't remind you (WCH12 Jan 2024).

A thermal decomposition must be endothermic: examiners expect you to notice a negative answer is wrong (WCH12 Jun 2019).

COMMON MISTAKE

A "negative temperature change" is never credited: describe a temperature **fall** and a positive ΔH (WCH12 Jan 2022).

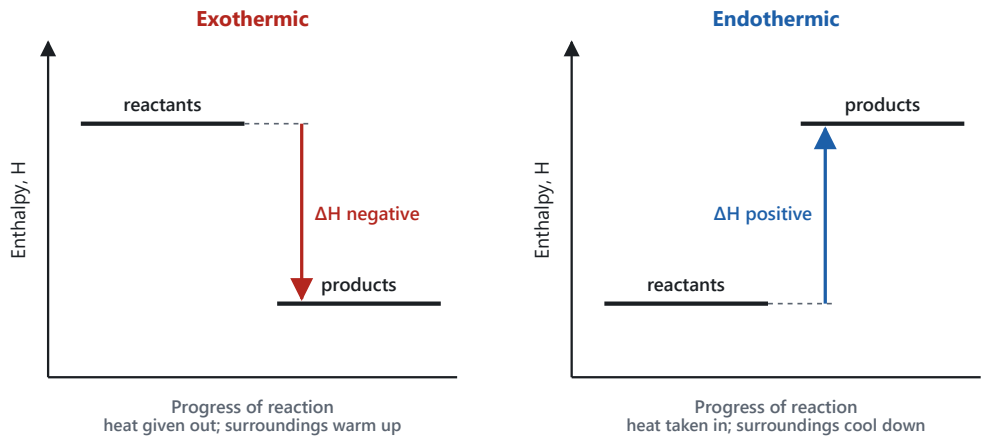
Saying neutralisation "requires energy". That implies it is endothermic; it releases energy (WCH12 Oct 2023 examiner report).

Dropping the minus sign on combustion and neutralisation enthalpies.

SPEC 6.3 Enthalpy level diagrams

WHAT THIS SPEC POINT ASKS

Draw and interpret enthalpy level diagrams for exothermic and endothermic changes.



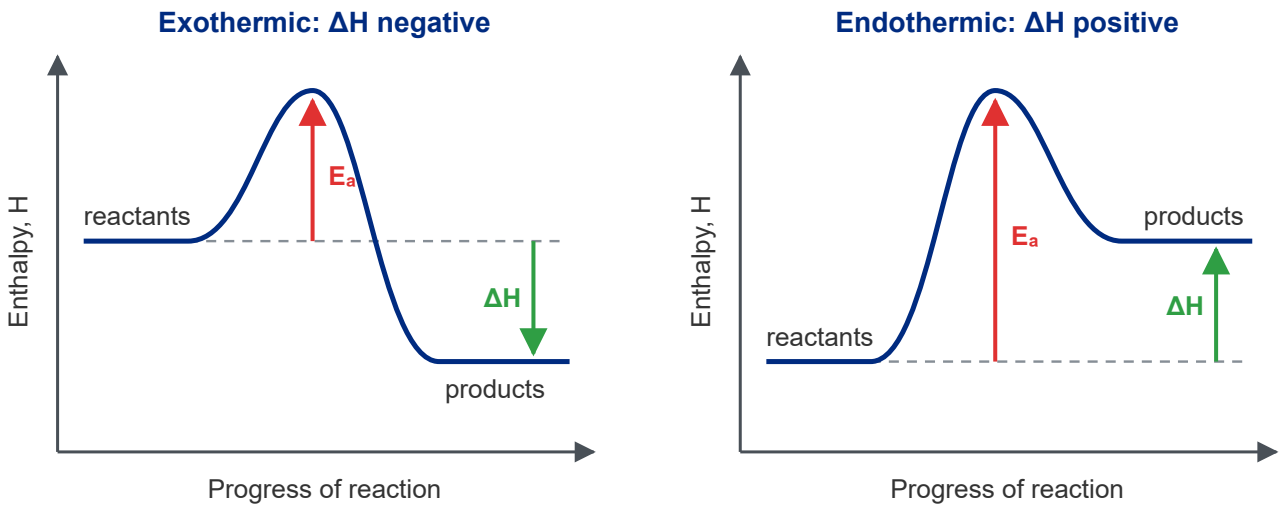
Exothermic: products below reactants, arrow points down. Endothermic: products above reactants, arrow points up.

How to draw one for full marks

1. Vertical axis labelled **enthalpy** (or H, or energy). Not "ΔH" or "enthalpy change".
2. Two horizontal lines: **reactants** and **products**. Label each, ideally with formulae and state symbols (e.g. $CH_4(g) + 2O_2(g)$).
3. Exothermic: products **lower**. Endothermic: products **higher**.
4. One **single-headed arrow** from the reactant level to the product level, labelled ΔH (with its value if you know it, e.g. $\Delta_c H = -890$ kJ mol^{-1}).

Level diagram or reaction profile?

- An **enthalpy level diagram** shows only the start and finish levels.
- A **reaction profile** also shows the hump: the **activation energy (E_a)** is measured from the reactants up to the top of the hump (the transition state). Catalysed profiles come in Topic 9 (9.6).
- For an endothermic reaction, E_a must be **larger** than ΔH, because the hump has to be above the products.



Reaction profiles. The activation energy is measured from the reactants up to the top of the hump. ΔH is the difference between the reactants and the products, whatever happens in between.

MARK-SCHEME WORDING

- "Y axis labelled **enthalpy**" (ACCEPT energy / H). DO NOT ACCEPT "ΔH", "enthalpy change" or "energy change" as the axis label.
- "Products line below reactants" (exothermic), both levels labelled, arrow **downward** labelled with the ΔH.
- DO NOT ACCEPT a double-headed arrow, a line with no arrowhead, or an arrow labelled −ΔH. A hump is ignored on a level diagram, and so is a missing x-axis label.
- Endothermic profile: "products higher than reactants", with ΔH and E_a both labelled (ΔH may be written as its value, e.g. +133).

Often asked as: draw the diagram for a combustion with $\Delta_c H$ labelled (WCH12 Jun 2022); draw an endothermic profile with ΔH and E_a (WCH12 Jan 2022).

COMMON MISTAKE

Drawing the familiar exothermic shape when the question gives a **positive** ΔH. Read the sign first (WCH12 Jan 2022 examiner report).

Arrows pointing the wrong way: each wrong arrow loses a mark. E_a arrows start at the **reactants** level, not the products.

SPEC 6.4 Standard enthalpy change definitions

WHAT THIS SPEC POINT ASKS

Know the definitions of the standard enthalpy changes of reaction, formation, combustion, neutralisation and atomisation.

The five definitions

All are measured under **standard conditions** (100 kPa, stated temperature, usually 298 K) with all substances in their **standard states**.

Change	Definition (learn the key words)	Example equation
Reaction, $\Delta_r H^{\ominus}$	enthalpy change when the **molar quantities shown in the equation** react	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, $\Delta_r H = -92$ kJ mol^{-1}

Change	Definition (learn the key words)	Example equation
Formation, $\Delta_f H^{\ominus}$	enthalpy change when **one mole of a compound** is formed **from its elements** in their standard states	$2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$
Combustion, $\Delta_c H^{\ominus}$	enthalpy change when **one mole of a substance** is **completely burned in oxygen**	$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
Neutralisation, $\Delta_{neut} H^{\ominus}$	enthalpy change when an acid reacts with an alkali to form **one mole of water**	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
Atomisation, $\Delta_{at} H^{\ominus}$	enthalpy change when **one mole of gaseous atoms** is formed from the element in its standard state	$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$; $C(s) \rightarrow C(g)$

Points that catch people out

- **Elements have $\Delta_f H = 0$.** Forming an element in its standard state from itself is no change. So $\Delta_f H$ of $O_2(g)$, C(graphite), Na(s) are all zero. (But $\Delta_f H$ of O_3 or of $Br_2(g)$ is not zero: they aren't standard states.)
- **Formation equations can need fractions** to make exactly one mole of compound: $\frac{1}{2}O_2$, $\frac{3}{2}H_2$. Remember H_2, O_2, N_2 and the halogens are **diatomic**, and include the oxygen already inside an alcohol when you balance the elements.
- **Combustion and neutralisation are always exothermic.** Atomisation is always endothermic (bonds or metallic bonding must be broken).
- **Atomisation of a diatomic element** makes one mole of atoms, so the equation uses **$\frac{1}{2}X_2$**. $\Delta_{at} H$ of chlorine = $\frac{1}{2} \times$ E(Cl–Cl) = $\frac{1}{2} \times 243 \approx +122$ kJ mol^{-1}.
- **One equation can fit two definitions.** $C(s) + O_2(g) \rightarrow CO_2(g)$ is both $\Delta_c H$ of carbon and $\Delta_f H$ of CO_2. $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ is both $\Delta_c H$ of hydrogen and $\Delta_f H$ of water.
- **Scaling.** If the equation is doubled, $\Delta_r H$ doubles: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ has $\Delta_r H = 2 \times (-286) = -572$ kJ mol^{-1}. But $\Delta_f H$ and $\Delta_c H$ are always **per mole** of the named substance.
- **Neutralisation** of any strong acid with any strong alkali gives about **−57 kJ mol^{-1}**, because the reaction is always $H^+ + OH^- \rightarrow H_2O$. Sulfuric acid + NaOH: the equation that matches the definition is $\frac{1}{2}H_2SO_4 + NaOH \rightarrow \frac{1}{2}Na_2SO_4 + H_2O$.

MARK-SCHEME WORDING

- Why $\Delta_f H$ of oxygen is zero: "oxygen is an **element** and (it is) in its **standard state**" (WCH12 Jan 2022).
- Formation: "**one mole** of compound", "from its **elements**", "in their **standard states**".
- Formation cycles: "balanced elements **with state symbols**", "inclusion of multiples".

Often asked as: write the equation for a named enthalpy change; explain why a value is zero; explain why a given equation is (or isn't) a formation reaction.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Writing the **ionic** equation for a named enthalpy change, with spectator ions cancelled, e.g. $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ for neutralisation (WCH12 Oct 2023).

Changes of state can appear as a step in a cycle: sublimation of iodine, $I_2(s) \rightarrow I_2(g)$, $\Delta H = +62$ kJ mol^{-1} (WCH12 Jan 2026).

COMMON MISTAKE

Writing "under standard conditions" when the mark is for "elements in their **standard states**" (WCH12 Jan 2024 examiner report).

Writing H and O instead of H_2 and O_2 on the elements line, leaving out state symbols, or forgetting the O already in an alcohol (WCH12 Jan 2022 examiner report).

Defining combustion "in air" or without "complete".

SPEC 6.5 Calculating enthalpy changes from experiments

WHAT THIS SPEC POINT ASKS

(i) Use energy transferred = mass × specific heat capacity × temperature change. (ii) Turn that into ΔH in kJ mol^{-1}. Limited to mixing in an insulated container and to combustion calorimetry.

The two equations

$$q = m\,c\,\Delta T$$

Symbol	Meaning	Unit
q	energy transferred to (or from) the surroundings	J
m	mass of the **water or solution being heated**, not the reagent	g
c	specific heat capacity of water, **4.18**	J g^{-1} $°C^{-1}$ (= J g^{-1} K^{-1})
ΔT	temperature change	°C (or K: a change of 1 °C = 1 K)

ΔH = −q ÷ n (with q in kJ), where n is the number of **moles of the substance the ΔH refers to**, usually the limiting reagent, or the water formed for neutralisation.

$$\Delta H = \frac{-q}{n}$$

- Put the **sign** in by thinking: temperature up → negative; temperature down → positive.
- For solutions, assume the **density is 1.00 g cm^{-3}**, so 50.0 cm^3 of solution has m = 50.0 g. Add the volumes of two solutions that are mixed.
- A solid added to a solution: m is still just the solution. Don't add the mass of the solid.

Method 1: mixing in an insulated container

- Use a **polystyrene cup** (a good insulator with a low heat capacity), with a lid, standing in a beaker for stability.
- Measure a known volume of one solution, record its temperature for a few minutes, add the second reagent (in **excess** if the other is the one you'll calculate with), stir, and record the temperature regularly.
- Used for neutralisation, displacement (e.g. Zn + $CuSO_4$), dissolving, and the reactions in Core Practical 2.

WORKED EXAMPLE

Worked example: enthalpy of neutralisation

25.0 cm^3 of 2.00 mol dm^{-3} HCl is mixed with 25.0 cm^3 of 2.00 mol dm^{-3} NaOH. The temperature rises by 13.5 °C.

q = m c ΔT = 50.0 × 4.18 × 13.5 = 2821.5 J = 2.8215 kJ

n(H_2O formed) = n(HCl) = 2.00 × 25.0 ÷ 1000 = 0.0500 mol

$\Delta_{neut}H$ = −2.8215 ÷ 0.0500 = **−56.4 kJ mol⁻¹** (3 s.f., negative because the temperature rose)

Worked example: neutralisation

1. 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid and 50.0 cm^3 of 1.00 mol dm^{-3} sodium hydroxide solution, both at 19.8 °C, are mixed. The extrapolated maximum temperature is 26.4 °C. Calculate the enthalpy change of neutralisation.

2. Both solutions end up warmer, so the mass is the total volume, 100.0 cm^3 = 100.0 g. ΔT = 26.4 − 19.8 = 6.6 K.

$$Q = 100.0 \times 4.18 \times 6.6 = 2758.8 \text{ J}$$

3. The equation is HCl + NaOH → NaCl + H_2O, so the moles of water formed equal the moles of acid (and of alkali):

$$n(\text{H}_2\text{O}) = 1.00 \times \frac{50.0}{1000} = 0.0500 \text{ mol}$$

4. $$\Delta_{\text{neut}} H = \frac{-2758.8}{0.0500} = -55\,176 \text{ J mol}^{-1} = -55.2 \text{ kJ mol}^{-1}$$

   Always finish with a sign, units and a sensible number of significant figures (normally 3, or fewer if the data have fewer).

Method 2: combustion calorimetry

- A **spirit burner** of the fuel heats a known mass of water in a **copper can** (copper conducts heat well) clamped above it.
- Weigh the burner (with its cap on, to stop evaporation) before and after; the loss in mass is the fuel burned.
- Stir the water and record its starting and highest temperatures. Use a draught shield, keep the flame close to the can, and use a lid on the can.

WORKED EXAMPLE

Worked example: enthalpy of combustion of ethanol

Burning 0.460 g of ethanol (M_r 46.0) heats 100.0 g of water by 13.2 °C.

q = 100.0 × 4.18 × 13.2 = 5517.6 J = 5.5176 kJ ← mass of the **water**, not the ethanol

n(ethanol) = 0.460 ÷ 46.0 = 0.0100 mol

Δ_cH = −5.5176 ÷ 0.0100 = **−552 kJ mol⁻¹**

The data-book value is −1367 kJ mol⁻¹. Why the experiment is so far off is 6.8.

Worked example: combustion of ethanol

1. A spirit burner containing ethanol, C_2H_5OH, loses 0.920 g while heating 200 g of water from 19.5 °C to 37.0 °C. Calculate Δ_cH for ethanol.

2 Energy absorbed by the water (ΔT = 17.5 K):

$$Q = 200 \times 4.18 \times 17.5 = 14\,630 \text{ J}$$

m is the mass of water in the can, not the mass of fuel burned.

3 Moles of ethanol burned, using M_r = 2(12.0) + 6(1.0) + 16.0 = 46.0:

$$n = \frac{0.920}{46.0} = 0.0200 \text{ mol}$$

4

$$\Delta_c H = \frac{-14\,630}{0.0200} = -731\,500 \text{ J mol}^{-1} = -732 \text{ kJ mol}^{-1}$$

The data book value is −1367 kJ mol^{-1}, so only about half of the energy released reached the water.

WORKED EXAMPLE

Worked example: working backwards to a temperature

A hand warmer reaction has $\Delta_r H$ = −90.0 kJ mol^{-1}. 0.0150 mol reacts in 50.0 g of water at 20.0 °C. What is the final temperature?

q = 0.0150 × 90.0 = 1.35 kJ = 1350 J

ΔT = q ÷ (m c) = 1350 ÷ (50.0 × 4.18) = 6.46 °C

Final temperature = 20.0 + 6.5 = **26.5 °C**

MARK-SCHEME WORDING

- Marks go: energy transferred (mcΔT) → moles → energy ÷ moles → **"sign and units and 2 or 3 s.f."**. Intermediate rounding is ignored.
- m is the mass of the **solution / water**: "50.0 × 4.18 × 5.2", not the salt (WCH12 Oct 2020). DO NOT AWARD 100.305 × 4.18 × 13.2 when 0.305 g of fuel was burned (WCH12 Jun 2022).
- c = 4.18 J g^{-1} °C^{-1} almost every time; 4.2 has been accepted where a paper used it.
- "Correct answer with no working scores full marks" is common, but show working so you can get transferred-error marks.

Often asked as: neutralisation, displacement or dissolving data in a cup; combustion of an alcohol; a thermometric titration; working back from ΔH to a temperature.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Thermometric titration: add acid in portions and plot temperature against volume. Two straight lines of best fit cross at the end point; the total mass heated includes the acid added (WCH12 Jun 2024).

Energy per gram or per cm³ of a fuel from $\Delta_c H$, M_r and density (WCH12 Jun 2022).

Percentage **efficiency** of a heater or engine = useful energy ÷ energy from combustion × 100 (WCH12 Jun 2023).

COMMON MISTAKE

Using the mass of the salt or fuel in mcΔT, or adding it to the mass of water (WCH12 Oct 2020, Oct 2022, Jan 2022 examiner reports).

Adding 273 to ΔT. A temperature **change** is the same in °C and K (WCH12 Oct 2021).

Forgetting to convert J to kJ, or to find the limiting reagent before working out n.

Using the M_r of a hydrated salt without its water of crystallisation (WCH12 Jan 2021).

SPEC 6.6 Hess's law and enthalpy cycles

WHAT THIS SPEC POINT ASKS

Know Hess's law and use it to (i) build enthalpy cycles and (ii) calculate enthalpy changes from given data, data from a table, or experimental results.

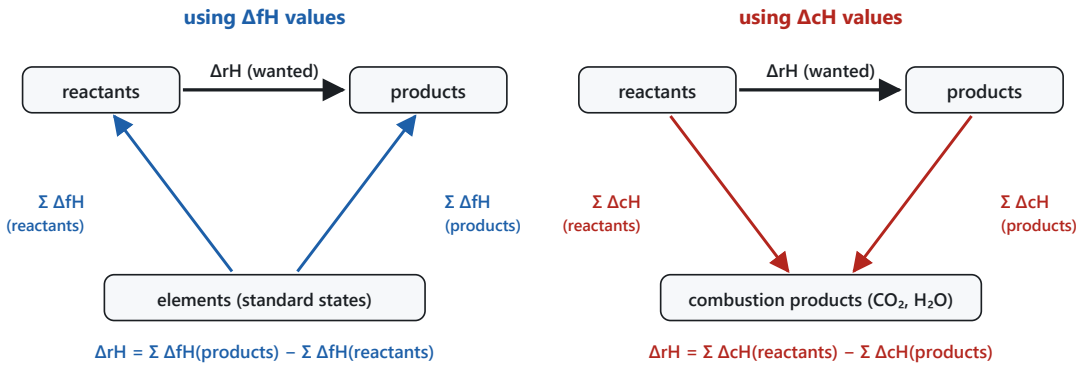
The law

KEY IDEA

Hess's law: the total enthalpy change for a reaction is independent of the route taken, as long as the starting and finishing conditions are the same. It follows from conservation of energy.

It lets you find ΔH for reactions you can't measure directly: reactions that are too slow, give side products, can't be started easily, or happen with a solid whose temperature you can't measure (e.g. $CaCO_3 \rightarrow CaO + CO_2$, or anhydrous $CuSO_4$ + water → hydrated $CuSO_4$).

The two standard cycles



With $\Delta_f H$ data the elements go at the bottom and the arrows point up. With $\Delta_c H$ data the combustion products go at the bottom and the arrows point down.

Route rule: going round the cycle the other way, follow the arrows; when you go against an arrow, **change its sign**. Multiply every value by the number of moles in the equation.

$$\Delta_r H = \sum \Delta_f H(\text{products}) - \sum \Delta_f H(\text{reactants}) \qquad \Delta_r H = \sum \Delta_c H(\text{reactants}) - \sum \Delta_c H(\text{products})$$

WORKED EXAMPLE

Worked example: $\Delta_r H$ from enthalpies of formation

Find $\Delta_r H$ for $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(l)$.

$\Delta_f H$ / kJ mol^{-1}: $NaHCO_3(s)$ −950.8; $Na_2CO_3(s)$ −1130.7; $CO_2(g)$ −393.5; $H_2O(l)$ −285.8

$\Sigma \Delta_f H$(products) = (−1130.7) + (−393.5) + (−285.8) = −1810.0

$\Sigma \Delta_f H$(reactants) = 2 × (−950.8) = −1901.6

$\Delta_r H$ = −1810.0 − (−1901.6) = **+91.6 kJ mol^{-1}** (positive, as a decomposition should be)

Worked example: reducing iron(III) oxide

1. Calculate $\Delta_r H$ for $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$. Data, $\Delta_f H$ / kJ mol^{-1}: $Fe_2O_3(s)$ −824.2, $CO(g)$ −110.5, $CO_2(g)$ −393.5.

2. Sum for the products. Iron is an element in its standard state, so its value is zero. Multiply each value by the number of moles in the equation:

$$\sum \Delta_f H(\text{products}) = 2(0) + 3(-393.5) = -1180.5 \text{ kJ mol}^{-1}$$

3. Sum for the reactants:

$$\sum \Delta_f H(\text{reactants}) = (-824.2) + 3(-110.5) = -1155.7 \text{ kJ mol}^{-1}$$

4. Products minus reactants:

$$\Delta_r H = -1180.5 - (-1155.7) = -24.8 \text{ kJ mol}^{-1}$$

Watch the double negative: subtracting −1155.7 is the same as adding 1155.7.

WORKED EXAMPLE

Worked example: $\Delta_f H$ from enthalpies of combustion

Find $\Delta_f H$ of ethanol: $2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$. This can't be done directly.

$\Delta_c H$ / kJ mol^{-1}: $C(s)$ −393.5; $H_2(g)$ −285.8; $C_2H_5OH(l)$ −1367.3

Cycle: elements at top left, ethanol at top right, $2CO_2 + 3H_2O$ at the bottom; all arrows point **down**.

$\Delta_f H = \Sigma \Delta_c H$(reactants) − $\Sigma \Delta_c H$(products)

= [2 × (−393.5) + 3 × (−285.8)] − (−1367.3)

= −1644.4 + 1367.3 = **−277.1 kJ mol^{-1}**

(Oxygen has no $\Delta_c H$: it is not a fuel.)

Worked example: the enthalpy of formation of propane

1. Calculate $\Delta_f H$ of propane, $3C(s) + 4H_2(g) \rightarrow C_3H_8(g)$, from $\Delta_c H$ / kJ mol^{-1}: $C(s)$ −393.5, $H_2(g)$ −285.8, $C_3H_8(g)$ −2219.2.

Carbon and hydrogen won't combine to give just propane, so this can't be measured directly. All three substances burn easily, though.

2. Burn the reactants (down the left arrow):

$$\sum \Delta_c H(\text{reactants}) = 3(-393.5) + 4(-285.8) = -2323.7 \text{ kJ mol}^{-1}$$

3. The product: one mole of propane, $\sum \Delta_c H(\text{products}) = -2219.2 \text{ kJ mol}^{-1}$.

4. Reactants minus products:

$$\Delta_f H(C_3H_8) = -2323.7 - (-2219.2) = -104.5 \text{ kJ mol}^{-1}$$

WORKED EXAMPLE

Worked example: a cycle from two experiments

$MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$ can't be measured directly. Instead react each solid with HCl:

ΔH_1: $MgCO_3(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + CO_2(g) + H_2O(l)$

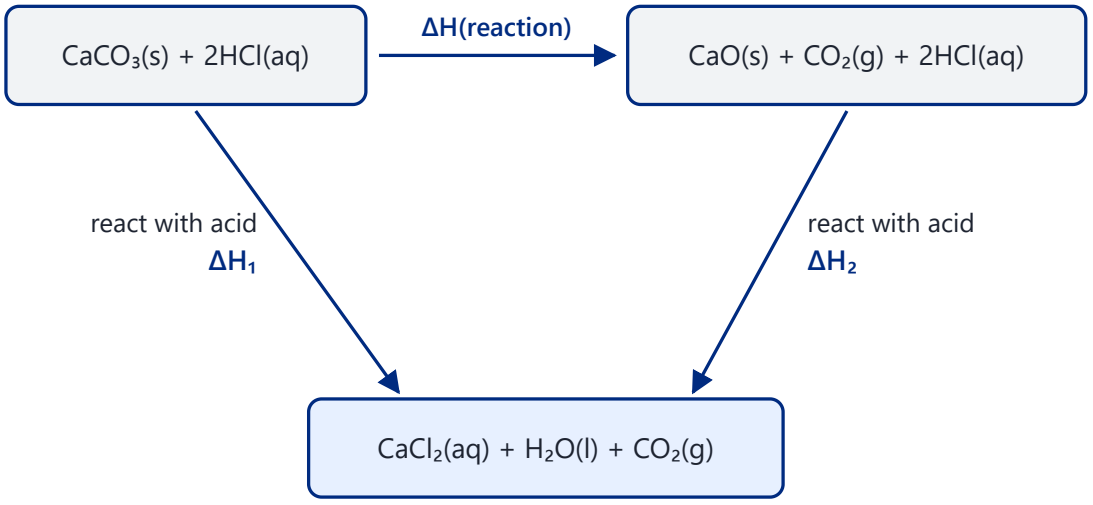
ΔH_2: $MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$

Cycle: $MgCO_3(s) + 2HCl(aq)$ on the top left, $MgO(s) + CO_2(g) + 2HCl(aq)$ on the top right, $MgCl_2(aq) + H_2O(l) + CO_2(g)$ at the bottom.

Going the long way round: down by ΔH_1, then **up against** the ΔH_2 arrow.

$\Delta_r H = \Delta H_1 - \Delta H_2$. With $\Delta H_1 = -50.3$ and $\Delta H_2 = -150.9$ kJ mol^{-1}:

$\Delta_r H = (-50.3) - (-150.9) =$ **+100.6 kJ mol^{-1}** (positive, as a decomposition should be)



Calcium carbonate and calcium oxide both react with hydrochloric acid to give the same products. Going round the cycle, $\Delta H(\text{reaction}) + \Delta H_2 = \Delta H_1$.

Combining equations (the algebra way)

You can also add and subtract equations like algebra. Reverse an equation → change the sign of its ΔH. Multiply an equation → multiply its ΔH. Cancel species that appear on both sides. Check that what's left is the target equation.

EXAM TIP

Put **brackets round every value** so the minus signs survive: "−(−1367.3)", not "−−1367.3". Then check the sign makes sense (decomposition positive, combustion negative).

MARK-SCHEME WORDING

- Cycle marks: "boxes / species correct with state symbols", "arrows correct", then "correct application and evaluation **including sign**".
- **No transferred error from an incorrect cycle**, but TE from an earlier numerical answer is allowed.
- Multiply by the coefficients: e.g. 2 × $\Delta_f H$(ethanoic acid) (WCH12 Jun 2021).

Often asked as: complete a formation or combustion cycle and calculate; a cycle built from two experiments ($CuSO_4$ hydration, $MgCO_3$ decomposition); combine three equations.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Cycles using the **enthalpy of solution** of an anhydrous and a hydrated salt to get the enthalpy of hydration (WCH12 Jan 2020, Oct 2024).

A cycle where the unknown is $\Delta_f H$ of one species, here CO from phosgene data: −220.1 − (−107.6) = −112.5 kJ mol^{-1} (WCH12 Jun 2021).

COMMON MISTAKE

Drawing the cycle the wrong way round, or getting the subtraction backwards (products − reactants for $\Delta_f H$; reactants − products for $\Delta_c H$) (WCH12 Jun 2019, Jan 2020 examiner reports).

Forgetting to multiply by the coefficients, or leaving a $\Delta_c H$ out of the sum (WCH12 Oct 2020 examiner report).

Copying values wrongly from the question into the working.

SPEC 6.7 Core Practical 2: an enthalpy change by Hess's law

WHAT THIS SPEC POINT ASKS

Find the enthalpy change of a reaction indirectly, using Hess's law and two measured enthalpy changes.

The idea

The target, e.g. the thermal decomposition $2KHCO_3(s) \rightarrow K_2CO_3(s) + CO_2(g) + H_2O(l)$, can't be measured: it needs strong heating, and you can't measure the temperature change of a solid being heated. Both solids react with hydrochloric acid at room temperature, though, and both give the **same products**, so a cycle links them.

- ΔH_1: $K_2CO_3(s) + 2HCl(aq) \rightarrow 2KCl(aq) + CO_2(g) + H_2O(l)$ (temperature **rises**)
- ΔH_2: $KHCO_3(s) + HCl(aq) \rightarrow KCl(aq) + CO_2(g) + H_2O(l)$ (temperature **falls**)
- Target: $\Delta_r H = 2\Delta H_2 - \Delta H_1$

CORE PRACTICAL

Core practical 2: method in brief

1. Measure **30.0 cm³ of 2 mol dm⁻³ HCl** (an excess) into a polystyrene cup standing in a beaker. Use a burette or measuring cylinder.
2. Weigh by difference about **3.5 g of anhydrous K_2CO_3** in a weighing bottle (record the exact mass).
3. Record the acid's temperature every 30 s for 2–3 minutes to get a steady starting value.
4. Add the solid at a known time, stir, and keep recording the temperature every 30 s for several minutes after the change stops. Add it **gradually** so the effervescence doesn't splash acid out.
5. Reweigh the empty bottle so you know exactly how much solid went in.
6. Repeat with **about 2.5 g of $KHCO_3$** and a fresh 30.0 cm³ of acid.
7. Plot each temperature–time graph and extrapolate to the time of mixing to find each ΔT (see 6.8).

Safety: eye protection; 2 mol dm⁻³ HCl is an irritant.

The same experiments can be run in the calorimetry simulation at the bottom of the page.

WORKED EXAMPLE

Worked example: processing the results

K_2CO_3: 3.45 g (M_r 138.2) → 0.0250 mol. ΔT = +6.6 °C.

$q_1 = 30.0 \times 4.18 \times 6.6 = 827.6$ J → $\Delta H_1 = -0.8276 \div 0.0250 = -33.1$ kJ mol⁻¹

$KHCO_3$: 2.50 g (M_r 100.1) → 0.0250 mol. ΔT = −6.3 °C.

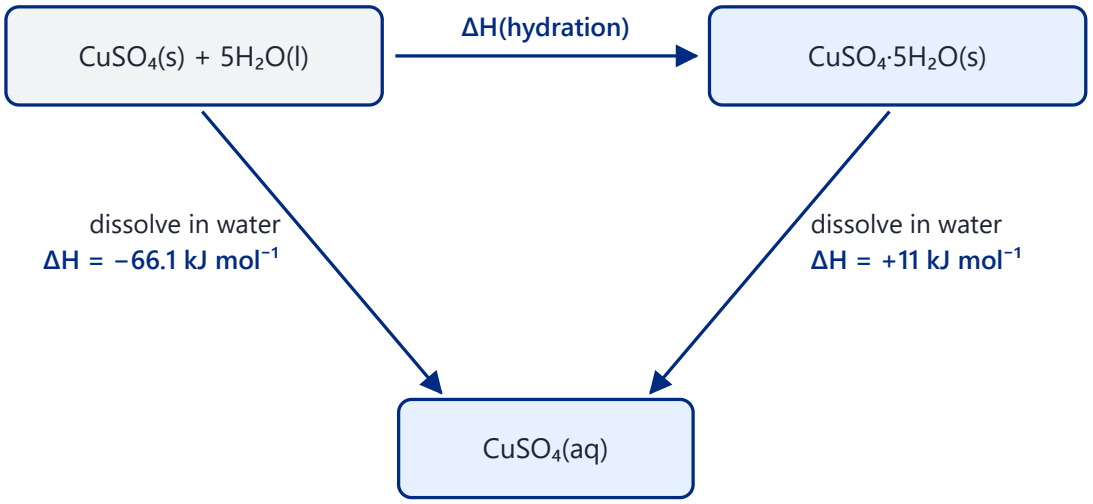
$q_2 = 30.0 \times 4.18 \times 6.3 = 790.0$ J → $\Delta H_2 = +0.7900 \div 0.0250 = +31.6$ kJ mol⁻¹

$\Delta_r H = 2\Delta H_2 - \Delta H_1 = 2(+31.6) - (-33.1) =$ **+96.3 kJ mol⁻¹**

Other versions the exam uses

The same "two experiments, one cycle" idea appears with:

- **Copper(II) sulfate**: dissolve anhydrous $CuSO_4$ and hydrated $CuSO_4 \cdot 5H_2O$ separately in water; ΔH(hydration) = ΔH(anhydrous dissolving) − ΔH(hydrated dissolving). To keep the final solutions the same, use less water with the hydrated salt, because its crystals already contain 5 mol of water per mole (WCH12 Jan 2020).
- **Magnesium sulfate**, $MgSO_4$ and $MgSO_4 \cdot 7H_2O$ (WCH12 Oct 2024).
- **MgO and $MgCO_3$** (or CaO and $CaCO_3$) with HCl, to find the carbonate's decomposition enthalpy (WCH12 Jun 2025).



Both salts dissolve to give $CuSO_4(aq)$. Following the arrows, ΔH(hydration) + 11 = −66.1, so ΔH(hydration) = −77.1 kJ mol^{-1}.

Worked example: Core Practical 2 results

1 Experiment 1: 3.99 g of anhydrous $CuSO_4$ (M_r = 159.6) dissolved in 50.0 g of water. Extrapolated ΔT = +7.6 K.

$$Q = 50.0 \times 4.18 \times 7.6 = 1588.4 \text{ J}, \qquad n = \frac{3.99}{159.6} = 0.0250 \text{ mol}$$

$$\Delta H_1 = \frac{-1588.4}{0.0250} = -63.5 \text{ kJ mol}^{-1}$$

2 Experiment 2: 6.24 g of $CuSO_4 \cdot 5H_2O$ (M_r = 249.6) dissolved in 50.0 g of water. Extrapolated ΔT = −1.3 K, so heat was taken in from the water.

$$Q = 50.0 \times 4.18 \times 1.3 = 271.7 \text{ J}, \qquad n = \frac{6.24}{249.6} = 0.0250 \text{ mol}$$

$$\Delta H_2 = \frac{+271.7}{0.0250} = +10.9 \text{ kJ mol}^{-1}$$

A temperature fall means an endothermic process, so this ΔH is positive.

3 Go round the cycle. Route 1: anhydrous salt + water → solution, ΔH_1. Route 2: anhydrous salt + water → hydrated salt (ΔH, unknown), then hydrated salt → solution (ΔH_2).

$$\Delta H + \Delta H_2 = \Delta H_1 \quad \Rightarrow \quad \Delta H = -63.5 - (+10.9) = -74.4 \text{ kJ mol}^{-1}$$

The data book value is about −78 kJ mol^{-1}. The biggest error is in ΔH_2: a fall of only 1.3 K has a large percentage uncertainty.

MARK-SCHEME WORDING

Why the direct reaction can't be measured:

- anhydrous salt + water: "adding water … will produce a solution and does not form a solid", "some of the water may turn to steam", "difficult to know when exactly 5 mol of water has been added"
- decomposition: "the solid has to be heated", "it is difficult to measure the temperature of a solid / powder while it is being heated" (WCH12 Jun 2025)

How water dissolves an ionic solid: "breaks the lattice" and "hydrates the ions / ion–dipole attractions". DO NOT ACCEPT hydrogen bonds to Cu^{2+}, or "molecules of copper sulfate" (WCH12 Jan 2020).

Often asked as: draw the cycle for the two reactions, calculate ΔH_1 and ΔH_2 from data, combine them, and explain why the target can't be measured directly.

COMMON MISTAKE

Not recognising the practical in an unfamiliar setting: the cycle should be routine (WCH12 Jan 2020 examiner report).

Using the mass of the solid in mcΔT (WCH12 Oct 2022 examiner report).

Forgetting the factor of 2 for $2KHCO_3$, or the sign of ΔH_2 (it is positive because the temperature fell).

SPEC 6.8 Evaluating calorimetry results

WHAT THIS SPEC POINT ASKS

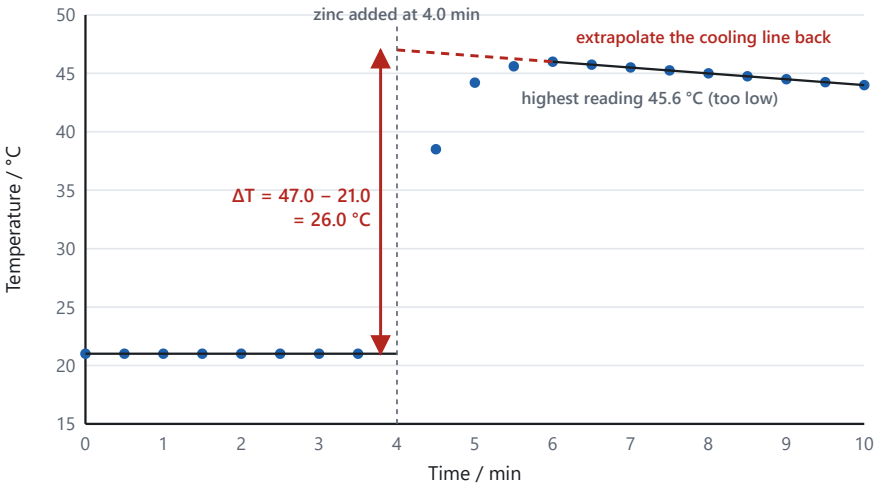
Evaluate results from mixing and combustion experiments: errors, uncertainties and assumptions. Draw suitable graphs and use cooling-curve corrections.

Assumptions behind the calculation

- The solution has the **density** and **specific heat capacity of water** (1.00 g cm^{-3}, 4.18 J g^{-1} $°C^{-1}$).
- **All the heat** goes into the water/solution: none is lost to the air or absorbed by the cup, thermometer or can.
- The reaction is **complete** (and, for combustion, the fuel burns **completely**).
- The other reagent is in excess, and the solid's own mass doesn't change the heat capacity.

The cooling-curve correction

The reaction doesn't happen instantly, so while the temperature is still rising, the mixture is **already losing heat** to the surroundings. The highest reading is therefore **lower** than the true value.



Draw two straight lines of best fit (before and after the reaction), extrapolate the cooling line back to the time of mixing and read ΔT there. Here ΔT = 26.0 °C, compared with only 24.6 °C from the highest reading.

1. Record the temperature for a few minutes **before** mixing to get a steady start temperature.
2. Mix at a recorded time (e.g. 4.0 min) without taking a reading at that moment.
3. Keep recording for several minutes after the maximum.
4. Draw a **straight line of best fit** through the cooling points and **extrapolate** it back to the time of mixing.
5. ΔT = extrapolated temperature − starting temperature at that time.

WORKED EXAMPLE

Worked example: from the graph

50.0 cm³ of 0.500 mol dm^{-3} $CuSO_4$ with excess zinc powder. ΔT from the graph = 26.0 °C.

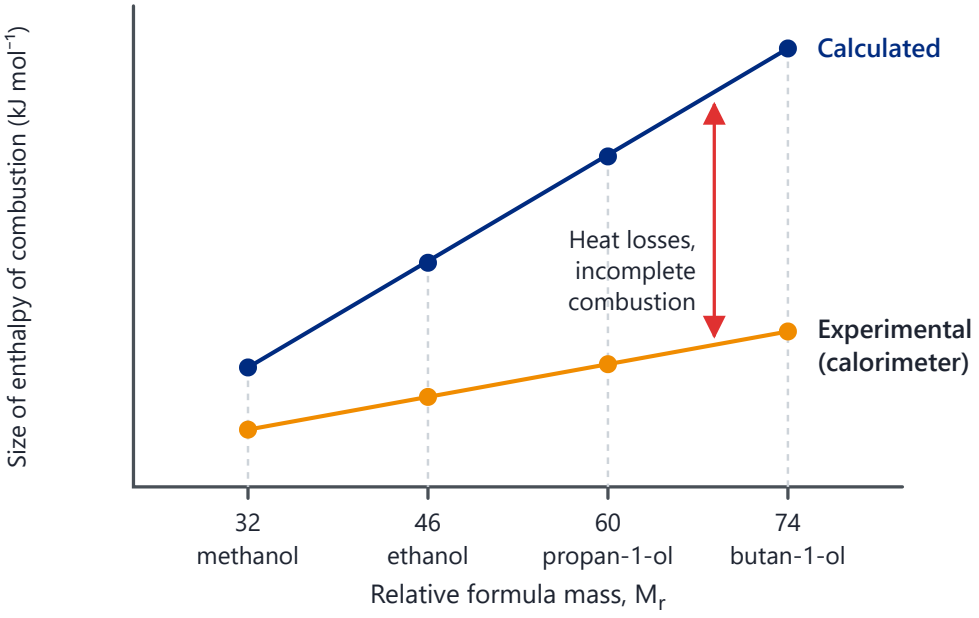
q = 50.0 × 4.18 × 26.0 = 5434 J; n($CuSO_4$) = 0.500 × 0.0500 = 0.0250 mol

$\Delta_r H$ = −5.434 ÷ 0.0250 = **−217 kJ mol^{-1}** for $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

Why combustion values come out too small

Measured $\Delta_c H$ values are always **less exothermic** than data-book values:

Source of error	Effect	Improvement
Heat lost to the surroundings (air, clamp)	less heat reaches the water	draught shield, lid, flame close to the can, insulate the can
Heat absorbed by the **copper can** and thermometer	not counted in mcΔT	include the can's heat capacity in the calculation
Incomplete combustion (soot, CO)	less energy released per mole	better air supply
Fuel evaporates from the wick	mass loss is more than the mass burned	keep the cap on when weighing; weigh quickly
Water in the products is a **gas**, not liquid as in the standard state	less energy released	(built into the method)



Sketch graph (not to scale) of the size of the enthalpy of combustion against M_r for the alcohols. Both sets of values rise steadily with each CH_2 group, but the calorimeter results are far smaller because of heat losses and incomplete combustion.

Uncertainty

- Percentage uncertainty = (uncertainty ÷ value measured) × 100.
- A **temperature change** (or a mass by difference, or a titre) comes from **two readings**, so double the reading uncertainty.

WORKED EXAMPLE

Worked example: percentage uncertainty in ΔT

Thermometer read to ±0.05 °C; ΔT = 13.2 °C.

% uncertainty = (2 × 0.05) ÷ 13.2 × 100 = **0.76%**

To reduce it: use a thermometer with finer graduations, or a bigger ΔT (more reagent, less water).

MARK-SCHEME WORDING

- Combustion errors (choose those the question allows): "**heat loss** (to the surroundings)", "**incomplete combustion**" (ACCEPT insufficient oxygen / soot / carbon monoxide), "some **ethanol evaporates**", "heat capacity of the beaker / can not taken into account" (WCH12 Jun 2022).
- IGNORE "not all the ethanol was burned", "thermometer", "impure ethanol", "not stirred", "no lid", "human error", "non-standard conditions".
- Graph: two straight lines of best fit extrapolated to the **time of mixing**; points covering at least half the grid; axes labelled with units.
- Percentage uncertainty: (2 × 0.05 × 100) ÷ 13.2 = 0.76%. DO NOT AWARD 0.75 or 0.80.

Often asked as: give two reasons the value is less exothermic than the data-book value; draw the lines and find ΔT; calculate a percentage uncertainty; compare two methods.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

When the stem rules out heat loss, the expected answers are incomplete combustion and evaporation of the fuel (WCH12 Oct 2021).

Comparing two methods: a calorimetric value is low because heat is lost or absorbed by the container; a gravimetric value is high if the product is wet or impure (WCH12 Oct 2020).

Reading the temperature at the stated time on a graph, and reading a maximum temperature then subtracting the start (WCH12 Jun 2024).

COMMON MISTAKE

Listing heat loss when the question says "other than heat loss" (WCH12 Oct 2021 examiner report).

Stock answers: "human error", "non-standard conditions", "the water changed state".

Comparing a data-book value with an experimental one when the question compares **two data sources** (WCH12 Jan 2025 examiner report).

Freehand or dotted lines of best fit, or a ΔT read from the highest point instead of the extrapolation.

SPEC 6.9 Bond enthalpy and mean bond enthalpy

WHAT THIS SPEC POINT ASKS

Understand bond enthalpy and mean bond enthalpy, use them to calculate enthalpy changes, and know the limitations of doing so.

Definitions

- **Bond enthalpy**: the enthalpy change to **break one mole of a particular bond** in a specific molecule, with everything in the **gas phase**. Always positive (breaking is endothermic).
- **Mean bond enthalpy**: the bond enthalpy for a type of bond **averaged over many different compounds**. The C–H bond in methane is not exactly the same strength as the C–H bonds in ethanol, so data books give an average.
- **Breaking bonds takes in energy (+). Making bonds gives out energy (−).**

The calculation

$\Delta_r H$ = Σ(bond enthalpies of bonds broken) − Σ(bond enthalpies of bonds made)

$$\Delta_r H = \sum(\text{bond enthalpies of bonds broken}) - \sum(\text{bond enthalpies of bonds made})$$

1. Draw **displayed formulae** of everything, so you can count every bond.
2. List bonds broken (reactants) with numbers, and bonds made (products).
3. Multiply each by its bond enthalpy; add up each side.
4. Broken − made. Negative → exothermic.

WORKED EXAMPLE

Worked example: combustion of ethanol from mean bond enthalpies

$C_2H_5OH(g) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$

Mean bond enthalpies / kJ mol^{-1}: C–H 413, C–C 347, C–O 358, O–H 464, O=O 498, C=O 805

Broken: 5 C–H (2065) + 1 C–C (347) + 1 C–O (358) + 1 O–H (464) + 3 O=O (1494) = **4728**

Made: 4 C=O (3220) + 6 O–H (2784) = **6004**

$\Delta_c H$ = 4728 − 6004 = **−1276 kJ mol⁻¹**

Note: 3 O_2, not 3.5. The alcohol already contains one O atom.

Worked example: burning methane, using mean bond enthalpies

1. Estimate ΔH for $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$. Draw out the molecules to count the bonds: methane has four C–H bonds, each O_2 has one O=O, CO_2 is O=C=O, and each water has two O–H bonds.

2. Bonds broken (endothermic):

$$4(\text{C–H}) + 2(\text{O=O}) = 4(413) + 2(498) = 1652 + 996 = +2648 \text{ kJ mol}^{-1}$$

3. Bonds made (exothermic):

$$2(\text{C=O}) + 4(\text{O–H}) = 2(805) + 4(464) = 1610 + 1856 = 3466 \text{ kJ mol}^{-1}$$

4. $$\Delta H = 2648 - 3466 = -818 \text{ kJ mol}^{-1}$$

Using enthalpies of formation for the same reaction (with water as a gas) gives −802 kJ mol^{-1}. The difference comes from using mean values instead of the actual bond enthalpies in these molecules.

Limitations

The bond-enthalpy answer (−1276) differs from the data-book $\Delta_c H$ (−1367) because:

1. **Mean values are averages** over many compounds, so they aren't exact for the bonds in this particular molecule.
2. Bond enthalpies apply to the **gas phase**, but in standard conditions ethanol and water are **liquids**. The enthalpy of vaporisation isn't included.

MARK-SCHEME WORDING

- Definition: "energy **required to break one mole** of a (specific) bond", "in the **gaseous** phase / state", "**averaged over a number of (different) compounds**". DO NOT ACCEPT "averaged over different types of bond"; IGNORE "mean" alone (WCH12 Oct 2021).
- Layout: "energy to break bonds = +…", "energy released making bonds = −…", "$\Delta_r H$ = broken − made".
- Limitation (two marks): "bond enthalpies are **averaged** over a (large) number of compounds" and "bond enthalpies refer to the **gas phase** but (octane / water) are **liquids**" (WCH12 Jun 2023). IGNORE standard conditions and experimental error answers.

Often asked as: combustion of an alcohol or alkane by bond enthalpies; define mean bond enthalpy; explain why the answer differs from the data-book value.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Each extra CH_2 in a homologous series means **one more C–C and two more C–H** broken, and **two more C=O and two more O–H** made, so $\Delta_c H$ changes by a nearly constant amount (WCH12 Jan 2026).

Counting bonds for many molecules: 8 SF_6 contain 48 S–F bonds (WCH12 Jan 2024).

COMMON MISTAKE

Doing made − broken (gives the right number with the wrong sign), or adding the two totals (WCH12 Jan 2025 examiner report).

Using 3.5 or 6.5 O_2 for an alcohol because you forgot its own oxygen (WCH12 Oct 2019).

Miscounting bonds: draw the displayed formula every time.

Giving "experimental error" as a limitation. There's no experiment: the answer is averages and gas phase (WCH12 Jun 2023 examiner report).

SPEC 6.10 Calculating mean bond enthalpies from enthalpy changes

WHAT THIS SPEC POINT ASKS

Calculate a mean bond enthalpy from enthalpy changes of reaction.

The approach

Turn the bond-enthalpy equation round: if ΔH and all the other bond enthalpies are known, the unknown one is the only gap. Often you build a cycle with **gaseous atoms** at the top.

WORKED EXAMPLE

Worked example: mean C–H bond enthalpy in methane

Data / kJ mol⁻¹: $\Delta_f H(CH_4, g) = -74.8$; $\Delta_{at}H(C) = +716.7$; $\Delta_{at}H(H) = +218.0$

Target: $CH_4(g) \rightarrow C(g) + 4H(g)$ (breaks all four C–H bonds)

Route: $CH_4 \rightarrow$ elements (reverse of formation, +74.8) → atoms (716.7 + 4 × 218.0)

$\Delta H = +74.8 + 716.7 + 872.0 = +1663.5$ kJ mol⁻¹

Mean C–H = 1663.5 ÷ 4 = **+416 kJ mol⁻¹**

WORKED EXAMPLE

Worked example: an unknown bond from $\Delta_r H$

$I_2(g) + Cl_2(g) \rightarrow 2ICl(g)$, $\Delta_r H = -30$ kJ mol⁻¹. E(I–I) = 151, E(Cl–Cl) = 243 kJ mol⁻¹.

Broken = 151 + 243 = 394. Made = 2 × E(I–Cl).

−30 = 394 − 2E(I–Cl) → 2E(I–Cl) = 424 → **E(I–Cl) = 212 kJ mol⁻¹**

Worked example: the N–H bond enthalpy in ammonia

1. $\Delta_f H(NH_3(g)) = -46.1$ kJ mol^{-1}. Use E(N≡N) = 945 and E(H–H) = 436 kJ mol^{-1} to find the N–H bond enthalpy. The formation equation is $\frac{1}{2}N_2(g) + 1\frac{1}{2}H_2(g) \rightarrow NH_3(g)$.

2. Bonds broken: half a mole of N≡N and one and a half moles of H–H:

$$\tfrac{1}{2}(945) + 1\tfrac{1}{2}(436) = 472.5 + 654 = 1126.5 \text{ kJ mol}^{-1}$$

3 Bonds made: three moles of N–H. Write the equation:

$$-46.1 = 1126.5 - 3E(\text{N–H})$$

4 Rearrange:

$$3E(\text{N–H}) = 1126.5 + 46.1 = 1172.6 \quad \Rightarrow \quad E(\text{N–H}) = 391 \text{ kJ mol}^{-1}$$

MARK-SCHEME WORDING

- Divide by the number of bonds: "O–H = 928 ÷ 2 = 464", "C–H = 1740 ÷ 4 = 435" (WCH12 Jun 2022).
- For I–Cl: "151 + 243 = 394"; DO NOT AWARD 2 × 151 + 2 × 243 (WCH12 Jan 2026).

Often asked as: find C–H, then O–H, then C–O and C–C step by step from atomisation data; find one unknown bond from $\Delta_r H$.

COMMON MISTAKE

Forgetting to divide by the number of bonds of that type in the molecule.

Using the wrong number of moles: 2ICl contains **two** I–Cl bonds.

SPEC 6.11 Bond enthalpy, which bond breaks and how fast

WHAT THIS SPEC POINT ASKS

Understand that bond enthalpy data hint at which bond breaks first in a reaction, how hard it is to break, and so how quickly a reaction goes at room temperature.

The idea

- To react, at least one bond usually has to break. The **weakest bond** (lowest bond enthalpy) tends to **break first**.
- A weak bond means a **lower activation energy**, so more collisions have enough energy and the reaction is **faster at room temperature**.
- A very strong bond means a high activation energy: the substance is **unreactive** even if the overall reaction would be exothermic.

Example	Bond enthalpy / kJ mol^{-1}	Consequence
C–Cl, C–Br, C–I in halogenoalkanes	346, 290, 228	iodoalkanes hydrolyse **fastest** (10.13)
N≡N in nitrogen	945	N_2 is very unreactive: the Haber process needs high temperature and a catalyst
F–F	158 (unusually weak)	fluorine reacts violently even in the cold
O=O vs C–H in a fuel	both strong	methane needs a spark to start burning

MARK-SCHEME WORDING

- "the **C–I bond is weaker** / has a lower bond enthalpy **than the C–Cl** bond", so it breaks more easily (WCH12 Jun 2022).
- Name the bond: never "iodine is weaker", never the H–I or H–Cl bond. DO NOT ACCEPT "C–I is easy to break" with no comparison. IGNORE bond length, electronegativity and size (WCH12 Oct 2024, Oct 2021).

Often asked as: why iodoalkanes react faster than chloroalkanes (the overlap with 10.13 is large); why N_2 is unreactive.

COMMON MISTAKE

Linking a **stronger** bond to a faster rate (WCH12 Jan 2022 examiner report).

Using polarity: C–Cl is the most polar but the **slowest** to hydrolyse. Bond strength wins.

Quick checks

QUICK CHECK

Which equation represents the standard enthalpy change of formation of ethanol?

- $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
- $2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$
- $4C(s) + 6H_2(g) + O_2(g) \rightarrow 2C_2H_5OH(l)$
- $2C(g) + 6H(g) + O(g) \rightarrow C_2H_5OH(l)$

Formation means one mole of the compound made from its elements in their standard states. The third option makes two moles, and the last one starts from gaseous atoms rather than the elements' standard states (C(s), $H_2(g)$, $O_2(g)$). The first is the combustion equation.

QUICK CHECK

25.0 cm^3 of 2.00 mol dm^{-3} HCl is mixed with 25.0 cm^3 of 2.00 mol dm^{-3} NaOH. The temperature rises by 13.0 °C. What is the enthalpy change of neutralisation?

- −27.2 kJ mol^{-1}
- −109 kJ mol^{-1}
- +54.3 kJ mol^{-1}
- −54.3 kJ mol^{-1}

The mass is the total of both solutions, 50.0 g, so Q = 50.0 × 4.18 × 13.0 = 2717 J. Moles of water = 2.00 × 0.0250 = 0.0500 mol, so ΔH = −2717 ÷ 0.0500 = −54 340 J mol^{-1} = −54.3 kJ mol^{-1}. Using only 25.0 g gives −27.2; it is negative because the temperature rose.

QUICK CHECK

$\Delta_c H$ of ethene is −1411 kJ mol^{-1} and $\Delta_c H$ of ethanol is −1367 kJ mol^{-1}. What is ΔH for $C_2H_4(g) + H_2O(l) \rightarrow C_2H_5OH(l)$?

- −44 kJ mol^{-1}
- +44 kJ mol^{-1}
- −2778 kJ mol^{-1}
- −330 kJ mol^{-1}

ΔH = $\Sigma\Delta_c H$(reactants) − $\Sigma\Delta_c H$(products) = (−1411 + 0) − (−1367) = −44 kJ mol^{-1}. Liquid water is already fully burned, so its enthalpy of combustion is zero. Subtracting the wrong way round gives +44.

QUICK CHECK

Using mean bond enthalpies H–H = 436, Cl–Cl = 243 and H–Cl = 432 kJ mol^{-1}, what is ΔH for $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$?

- +247 kJ mol^{-1}
- +185 kJ mol^{-1}
- −185 kJ mol^{-1}
- −1111 kJ mol^{-1}

Bonds broken = 436 + 243 = 679; bonds made = 2 × 432 = 864. ΔH = 679 − 864 = −185 kJ mol^{-1}. Forgetting that two moles of H–Cl form gives +247.

QUICK CHECK

Why does a calculation with mean bond enthalpies usually give a slightly different ΔH from a Hess cycle using enthalpies of formation?

- Bond breaking is exothermic, so the signs are reversed.
- Heat is lost to the surroundings during the calculation.
- Hess's law does not apply to gases.
- Mean values are averages over many compounds, not the actual values in these molecules.

Each mean bond enthalpy is averaged over many different molecules, so it is only approximately right for any particular molecule. (Bond breaking is endothermic, and heat loss affects experiments, not calculations.)

QUICK CHECK

Which of these enthalpy changes is always positive?

- Standard enthalpy change of combustion
- Standard enthalpy change of atomisation
- Standard enthalpy change of neutralisation
- Standard enthalpy change of formation

Making gaseous atoms from an element always means breaking bonds, which needs energy. Combustion and neutralisation are always exothermic. Formation is usually exothermic, but some compounds, such as ethene (+52 kJ mol^{-1}), have positive values.

QUICK CHECK

A student measures $\Delta_c H$ of propan-1-ol with a spirit burner and a copper can and gets −1150 kJ mol^{-1}. The data book value is −2021 kJ mol^{-1}. Which is the main reason for the difference?

- The specific heat capacity of water is too large.
- The water in the can was not stirred.
- Much of the heat from the flame escaped to the air instead of heating the water.
- The propan-1-ol contained too much oxygen.

Heat loss from the flame is by far the largest error in combustion calorimetry, with incomplete combustion next. Both make the measured value much less exothermic than the true value.